

AI Intelligent Freight Quote Generation System

Milestone 4 — Company Selection, Verification & Booking Workflow

M4 Objective

M4 starts after AI-generated company quote options are available. The customer selects a preferred freight company, the selected company's authorized agent verifies the shipment and commercial details, and the workflow handles approval, modification, rejection and final booking confirmation.

Where M4 Fits

Milestone	Purpose	Output
M1	Shipment creation + route + base pricing	Shipment request
M2	AI/ML price prediction	Intelligent pricing
M3	Weather/customs/route risk	Risk-aware quote
M4	Company selection + verification + booking	Verified booking

M1 → M2 → M3 → M4

CREATE SHIPMENT → AI PRICE → AI RISK → MULTIPLE COMPANY QUOTES → CUSTOMER SELECTS → COMPANY AGENT VERIFIES → APPROVE / MODIFY / REJECT → BOOKING

1. What Exactly Is Milestone 4?

M4 is the business confirmation layer. M1–M3 analyze and generate quote options. M4 converts the customer's preferred quote into a real operational request that the selected freight company must verify.

Step	Example
Shipment	Chennai → Rotterdam, 5,000 KG Electronics
AI Options	Company A ■85K • Company B ■80K • Company C ■92K
Customer Choice	Customer selects Company B
M4 Starts	Request goes only to Company B agent
Agent Decision	Approve / Modify / Reject / Request Info
Result	Confirmed booking or alternate action

Important: Quote selection is not booking confirmation. The selected company must first verify capacity, route, documents and final commercial terms.

2. Roles and Responsibilities in M4

Role	M4 Responsibility	Access
Customer	Select quote, review revision, upload documents, accept/reject revised offer	Own shipments
Company Agent	Verify availability, cargo, route, documents and price	Only own company
Company Manager	Optional approval for special/high-value cases	Own company
Platform Admin	Manage companies, agents, rules and monitor bookings	All platform
AI Services	Provide price/risk context and recommendations	Backend intelligence

Company Data Isolation Rule

If a customer selects Company B, only authorized Company B agents can access that verification request. Company A and C agents must never see it.

3. Complete M4 End-to-End Process

1. Quote Options Available

M1–M3 completed shipment, pricing and risk analysis.

2. Customer Compares

Customer compares price, ETA, risk and service.

3. Customer Selects

Customer chooses one company quote.

4. Selection Recorded

System stores shipment, quote and selected company IDs.

5. Status Updated

Status becomes PENDING_COMPANY_VERIFICATION.

6. Agent Notified

Selected company's authorized agent gets the request.

7. Agent Opens Request

Views shipment, quote, AI price/risk and documents.

8. Operational Verification

Checks capacity, route, container/vehicle and schedule.

9. Commercial Verification

Checks quote validity, price and surcharges.

10. Document Verification

Checks required shipment/customs documents.

11. Agent Decision

Approve, Modify, Reject or Request More Information.

12. Customer Notification

System sends decision to customer.

13. Customer Action

Accept revision, reject revision or upload missing data.

14. Final Confirmation

Approved request becomes a confirmed booking.

Master Process

QUOTE OPTIONS → CUSTOMER SELECTS → SELECTED COMPANY → AGENT NOTIFIED → VERIFICATION →
APPROVE / MODIFY / REJECT / INFO → CUSTOMER RESPONSE → BOOKING

4. M4 Decision Flow

CUSTOMER SELECTS QUOTE

↓

PENDING COMPANY VERIFICATION

↓

COMPANY AGENT REVIEWS

↓

APPROVE → BOOKING CONFIRMED

MODIFY → CUSTOMER REVIEWS → ACCEPT → BOOKING

REJECT → CUSTOMER SELECTS ANOTHER QUOTE

REQUEST INFO → CUSTOMER UPLOADS DETAILS → AGENT REVIEWS AGAIN

Why Can an Agent Modify the Quote?

Actual freight cost may change because of fuel surcharge, capacity shortage, seasonal demand, port charges, container availability or changes in shipment details. Every modification should have a reason and customer approval.

5. Company Agent Verification Checklist

Area	Agent Checks	Result
Shipment	Origin, destination, cargo, date	Valid / Issue
Cargo	Weight, volume, special handling	Accepted / Restriction
Capacity	Container/vehicle/space	Available / Not
Route	Operational feasibility	Approved / Alternative
Schedule	Pickup and ETA	Confirmed / Changed
Documents	Invoice, packing list, requirements	Complete / Missing
Commercial	Base rate, fuel, handling charges	Confirmed / Revised
Risk Context	M3 weather/customs/route alerts	Accepted / Attention
Quote Validity	Expiry date	Valid / Expired

Agent Actions

Action	When	Next Status
Approve	Everything valid	APPROVED
Modify	Price/ETA/terms changed	REVISION_PENDING_CUSTOMER
Reject	Cannot fulfill shipment	REJECTED
Request Info	Details/documents missing	AWAITING_CUSTOMER_INFO
Escalate	Special approval required	ESCALATED

6. M4 Dashboard Pages

Customer Portal

Page	Purpose
Compare Quotes	Compare company options
Quote Details	Price breakdown, ETA and risk
Selected Quote	Track verification
Revision Review	Accept/reject modified quote
Bookings	View final booking
Documents	Upload requested files

Company Agent Portal

Page	Purpose
Dashboard	New/pending/approved counts
Incoming Requests	Customer selections
Pending Verification	Requests requiring action
Shipment Details	Complete shipment context
Quote Verification	Price/commercial validation
Document Review	Verify documents
Booking Management	Confirmed bookings
Rejected Requests	History and reasons
Notifications	Workflow alerts

Platform Admin

Page	Purpose
Companies	Manage freight providers
Company Agents	Map users to companies
Selected Quotes	Monitor customer selections
Booking Requests	Track workflow
Verification Monitor	Monitor pending actions
Company Performance	Approval/rejection/response metrics
Reports	Conversion analytics
Audit Logs	Track all critical actions

7. Recommended Side Navigation

Customer	Company Agent	Platform Admin
Dashboard Create Shipment My Shipments Compare Quotes Selected Quotes Bookings Documents Notifications Profile	Dashboard Incoming Requests Pending Verification Shipment Requests Quote Verification Document Review Booking Management Approved Rejected Notifications	Dashboard Companies Company Agents Customers All Shipments All Quotes Selected Quotes Booking Requests Verification Monitor Performance Reports Audit Logs Settings

M4 Dashboard KPI Cards

Customer	Company Agent	Admin
Selected Quotes Pending Verification Revisions Pending Confirmed Bookings	New Requests Pending Verification Approved Today Rejected Today Response Time	Total Companies Selected Quotes Pending Reviews Confirmed Bookings Conversion Rate

8. M4 Data Models / Schemas Needed

Model	Purpose	Key Fields
FreightCompany	Freight provider data	companyId, name, modes, status
CompanyAgent	Maps agent to company	userId, companyId, role
CompanyQuote	Company-specific quote	shipmentId, companyId, price, ETA
QuoteSelection	Customer selected quote	quoteId, shipmentId, customerId, status
VerificationRequest	Agent verification workflow	selectionId, agentId, status
QuoteRevision	Modified commercial offer	originalQuoteId, newPrice, reason
Booking	Final booking	shipmentId, companyId, reference, status
VerificationChecklist	Verification records	requestId, checks, remarks
StatusHistory	Audit of state changes	oldStatus, newStatus, changedBy
Notification	Workflow alerts	userId, entityId, message

Core Relationship

ONE SHIPMENT → MANY COMPANY QUOTES → CUSTOMER SELECTS ONE → ONE VERIFICATION REQUEST → APPROVE / MODIFY / REJECT → ONE FINAL BOOKING

9. Core Schema Relationship



Relationship	Rule
Shipment → Company Quotes	One shipment can have multiple options
Selection → Company	Selected quote belongs to one company
Company → Agents	One company can have many agents
Selection → Revisions	Keep revision history
Approved Selection → Booking	One approved selection creates final booking

10. Suggested Backend / API Modules

Module	Responsibilities
Company Management	Create/update/activate companies
Agent Management	Create agents and map company access
Quote Selection	Select/cancel/get selected quote
Verification Workflow	Pending queue, assignment, checklist, decision
Quote Revision	Create revision and customer acceptance
Booking Management	Create, confirm, cancel, track booking
Documents	Request/upload/verify documents
Notifications	Agent/customer workflow alerts
Audit	Record all decisions and status changes
Analytics	Response time, approval rate, conversion

Example API Flow

POST selection → create verification request → notify selected company → agent gets pending request → submit verification decision → notify customer → customer accepts revision if needed → create booking → confirm booking

11. M4 Status Lifecycle

Stage	Status	Meaning
Quote	QUOTE_OPTIONS_AVAILABLE	Options ready
Customer	QUOTE_SELECTED	Customer selected one
System	PENDING_COMPANY_VERIFICATION	Waiting for company
Agent	UNDER_VERIFICATION	Agent reviewing
Agent	AWAITING_CUSTOMER_INFO	More details needed
Agent	REVISION_PENDING_CUSTOMER	Modified offer waiting
Agent	APPROVED	Company accepts
Agent	REJECTED	Company cannot fulfill
Customer	REVISION_ACCEPTED	Customer accepts modification
Booking	BOOKING_CONFIRMED	Final booking
Booking	BOOKING_CANCELLED	Cancelled
Alternative	RESELECT_QUOTE	Choose another option

Happy Path

QUOTE_OPTIONS_AVAILABLE → QUOTE_SELECTED → PENDING_COMPANY_VERIFICATION → UNDER_VERIFICATION → APPROVED → BOOKING_CONFIRMED

Revision Path

QUOTE_SELECTED → UNDER_VERIFICATION → REVISION_PENDING_CUSTOMER → REVISION_ACCEPTED → APPROVED → BOOKING_CONFIRMED

12. Detailed Sequence Flow

#	Actor	Action	Result
1	Customer	Views quotes	Company options shown
2	Customer	Selects Company B	Selection recorded
3	System	Creates snapshot	Selected price/terms preserved
4	System	Creates verification	Pending request
5	System	Notifies Company B	Only B agents see it
6	Agent	Reviews request	Shipment + quote + risk
7	Agent	Checks availability	Checklist updated
8	Agent	Checks documents	Missing items flagged
9	Agent	Checks price	Confirmed or revised
10	Agent	Submits decision	Workflow updated
11	System	Records audit	Status history saved
12	System	Notifies customer	Decision visible
13	Customer	Responds	Accept/reject/upload
14	System	Creates booking	Booking reference
15	Customer	Views booking	Confirmed status

13. Sample M4 Test Data

Field	Value
Shipment ID	SHP-1001
Customer	ABC Electronics Pvt Ltd
Origin	Chennai, India
Destination	Rotterdam, Netherlands
Cargo	Electronics
Weight	5,000 KG
Mode	Sea Freight
Container	40 FT

Quote	Company	Price	ETA	Risk
QT-5001	OceanLink Logistics	■85,000	24 Days	Low
QT-5002	GlobalSea Freight	■80,000	28 Days	Medium
QT-5003	FastRoute Cargo	■92,000	20 Days	Low

Selected Quote Scenario	Value
Customer Selection	QT-5002 — GlobalSea Freight
Status	PENDING_COMPANY_VERIFICATION
Assigned Agent	AGT-204
Availability	Available
Documents	Complete
Price Revision	■80,000 → ■82,500
Reason	Fuel surcharge increase
Customer Decision	Accepted
Final Result	BOOKING_CONFIRMED
Booking Reference	BK-2026-10045

14. M4 Test Scenarios

#	Test Scenario	Expected Result
1	Customer selects valid quote	Correct selection/company stored
2	Expired quote selected	Blocked or refresh required
3	Company A accesses Company B request	Access denied
4	Agent approves request	Booking workflow starts
5	Agent modifies price	Revision created; original retained
6	Customer accepts revision	Booking proceeds
7	Customer rejects revision	Alternate quote allowed
8	Agent rejects no capacity	Reason stored + notification
9	Document missing	AWAITING_CUSTOMER_INFO
10	Customer uploads document	Returns to verification
11	Duplicate decision	Duplicate prevention works
12	Unauthorized booking change	Denied + audit
13	Notification fails	Workflow saved; retry
14	Agent response delayed	SLA alert
15	Booking confirmed	Reference visible to customer

15. Important Business Rules

Rule	Requirement
Company Isolation	Agents see only their company data
Quote Snapshot	Preserve original selected values
No Silent Changes	Every modification needs reason
Revision History	Never overwrite original quote
Approval Authority	Only authorized roles approve
Audit Trail	Track selection, verification and booking
Expiry	Expired quote cannot be confirmed directly
Customer Consent	Revised terms need acceptance
Document Privacy	Controlled access to sensitive files
Fallback	Rejected request allows another quote
AI Role	AI recommends; authorized human/company gives final operational approval

M4 Success Definition: A selected quote becomes a real booking only after required company verification and customer confirmation are completed.

16. M4 Implementation Checklist

Layer	Tasks
Customer Frontend	Compare, select, track verification, review revision, confirm booking
Agent Frontend	Incoming queue, shipment detail, checklist, approve/modify/reject
Admin Frontend	Companies, agents, booking monitor, analytics
Backend	Selection, verification, revision, booking, notification, audit
Database	Company, agent, selection, verification, revision, booking models
Security	RBAC + company-level data isolation
Integration	Connect M1 shipment + M2 price + M3 risk
Testing	Access, workflow, revision, rejection, documents, notifications
Analytics	Conversion, approval/rejection, response time

Recommended Development Order

1. Company Models → 2. Agent Mapping → 3. Company Quotes → 4. Customer Selection → 5. Verification Request → 6. Agent Dashboard → 7. Approve/Modify/Reject → 8. Customer Revision → 9. Booking → 10. Notifications + Audit + Testing

17. Final M4 Architecture Summary

M1 SHIPMENT
↓
M2 AI PRICE + M3 RISK
↓
MULTIPLE COMPANY QUOTES
↓
CUSTOMER COMPARES AND SELECTS ONE
↓
QUOTE SELECTION
↓
SELECTED COMPANY AGENT
↓
OPERATIONAL + DOCUMENT + COMMERCIAL VERIFICATION
↓
APPROVE → BOOKING CONFIRMED
MODIFY → CUSTOMER REVIEW → ACCEPT → BOOKING
REJECT → CUSTOMER SELECTS ANOTHER OPTION
REQUEST INFO → CUSTOMER UPLOADS → VERIFY AGAIN

One-Line Understanding

M4 converts an AI-generated freight quote into a verified real-world booking by connecting the customer's selected company with that company's authorized verification agent.

M4 Adds Real Business Value

Before M4	After M4
AI recommends prices	Customer selects a provider
Quote is an estimate	Company verifies commercial validity
Risk is analyzed	Operational capacity is checked
Proposal only	Booking can be confirmed
No company workflow	Company agents participate
Project ends at quote	Project reaches real business confirmation